



2025 SEPTEMBER

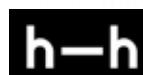
LEBANON

INFORMING HUMANITARIAN RESPONSE: A GUIDE TO REMOTE
SENSING ANALYSES OF CONFLICT DAMAGE



CONFLICT
PROGRAM

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TERMS & CONDITIONS

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GLOSSARY

AMPLITUDE

The strength or intensity of a (radar) signal, used in remote sensing to describe the magnitude of the returned wave.

BACKSCATTERING ANALYSIS

The study of how radar signals bounce back from surfaces, helping identify materials, textures, or damage.

CONFLICT DAMAGE

The physical destruction of buildings, infrastructure, or land due to armed conflict.

DEEP LEARNING

A type of machine learning using neural networks with many layers to recognize complex patterns in data, such as images.

EARTH OBSERVATION

Acquisition and analysis of data from satellites, aircraft, or sensors to monitor Earth's environment, resources, and human activities.

EXPERT VISUAL INTERPRETATION

Manual analysis of satellite or aerial images by trained specialists to identify features or changes.

MICROSOFT BUILDINGS

A dataset created by Microsoft using Machine Learning to map building footprints worldwide.

OPENSTREETMAP (OSM)

A collaborative, open-source map of the world created and maintained by volunteers.

OPTICAL

Refers to sensors that use visible and near-infrared light to capture images, similar to a camera.

OVERTURE

Collaboration of technology companies to compile open spatial datasets into a free, standardized, and interoperable database.

PHASE

The position of a radar wave relative to its cycle, used in interferometry to detect small surface movements.

REMOTE SENSING

The process of gathering information about the Earth without direct contact, typically using satellites or aircraft.

SAR (SYNTHETIC APERTURE RADAR)

A radar imaging technique that creates detailed images of the Earth's surface regardless of weather or light conditions.

INSAR (INTERFEROMETRIC SYNTHETIC APERTURE RADAR)

A radar technique that measures ground deformation by comparing phase differences between satellite passes.

SATELLITE IMAGERY

Pictures of Earth captured by satellites, available in various resolutions and spectral bands for analysis.



Aerial view of buildings in Beirut, Lebanon. This image was captured by iam hogir | [View on Pexels.](#)

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By December 2024, several conflict damage assessments were conducted in Lebanon, producing varying estimates of building damage. These differences highlight the need for stronger coordination and clearer understanding of remote methods.

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Remote damage assessments are increasingly used to document conflict impacts where direct access is limited. Navigating differences in methodology and results is essential for responsible use of damage data.

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DIVERGENCE IN FINDINGS: A COMPARISON OF REMOTE DAMAGE ASSESSMENTS IN LEBANON 2024

Remote conflict damage assessments in Lebanon as of December 2024 differed in their estimated building damage counts by up to a factor of three due to different approaches.

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UNDERSTANDING DIFFERENT APPROACHES FOR CONFLICT DAMAGE MAPPING

Remote mapping methods rely on satellite imagery, analytical techniques, and frequency of observation. Each choice influences accuracy and shapes reported outcomes.

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DISCUSSION AND CONCLUSION

Divergent results can either undermine or reinforce confidence in conflict damage analysis. Greater transparency, research, and coordination are needed for responsible use of assessments.

EXECUTIVE SUMMARY

As conflicts events increase globally ([ACLED Conflict Index](#)), reliable and accessible information on infrastructure damage has become critical to understand conflict impacts on communities. However, damage analysis results can differ based on methodology. Yet, there is currently no coordinated, focused effort to compare satellite-derived damage assessments, reduce duplication, or identify the most suitable methods for generating damage analysis for a given purpose.

There were at least seven different remote assessments of building conflict damage in southern Lebanon from October to December of 2024. **While each group used different methodologies and data sources, findings varied significantly.** Five of the seven shared actual counts of damaged buildings that were comparable across assessments. Looking at Bint Jbeil District damaged building counts as a case study; three assessments produced conservative assessments in high agreement with each other, and two estimated roughly three times the damage compared to the three more conservative estimates (see graph below).

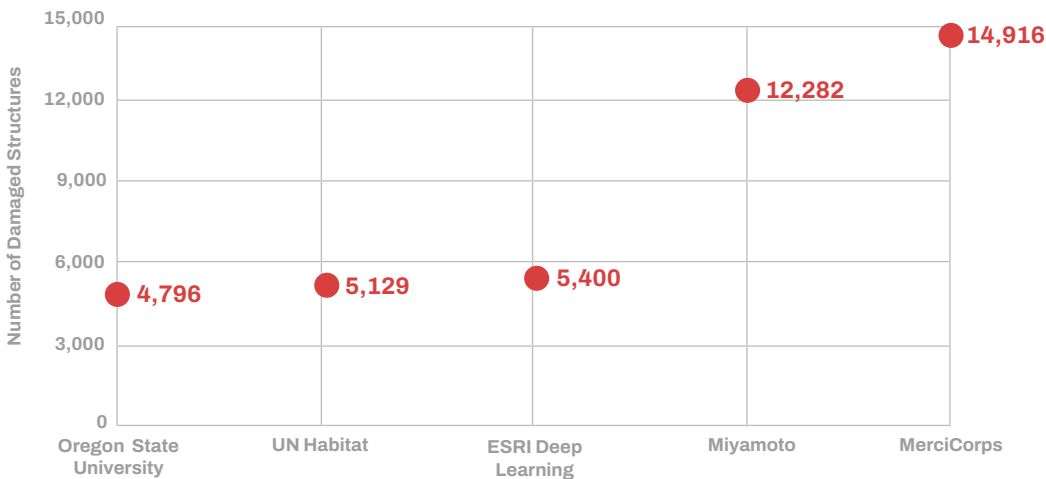
Key factors in each approach that can contribute to potential discrepancies:

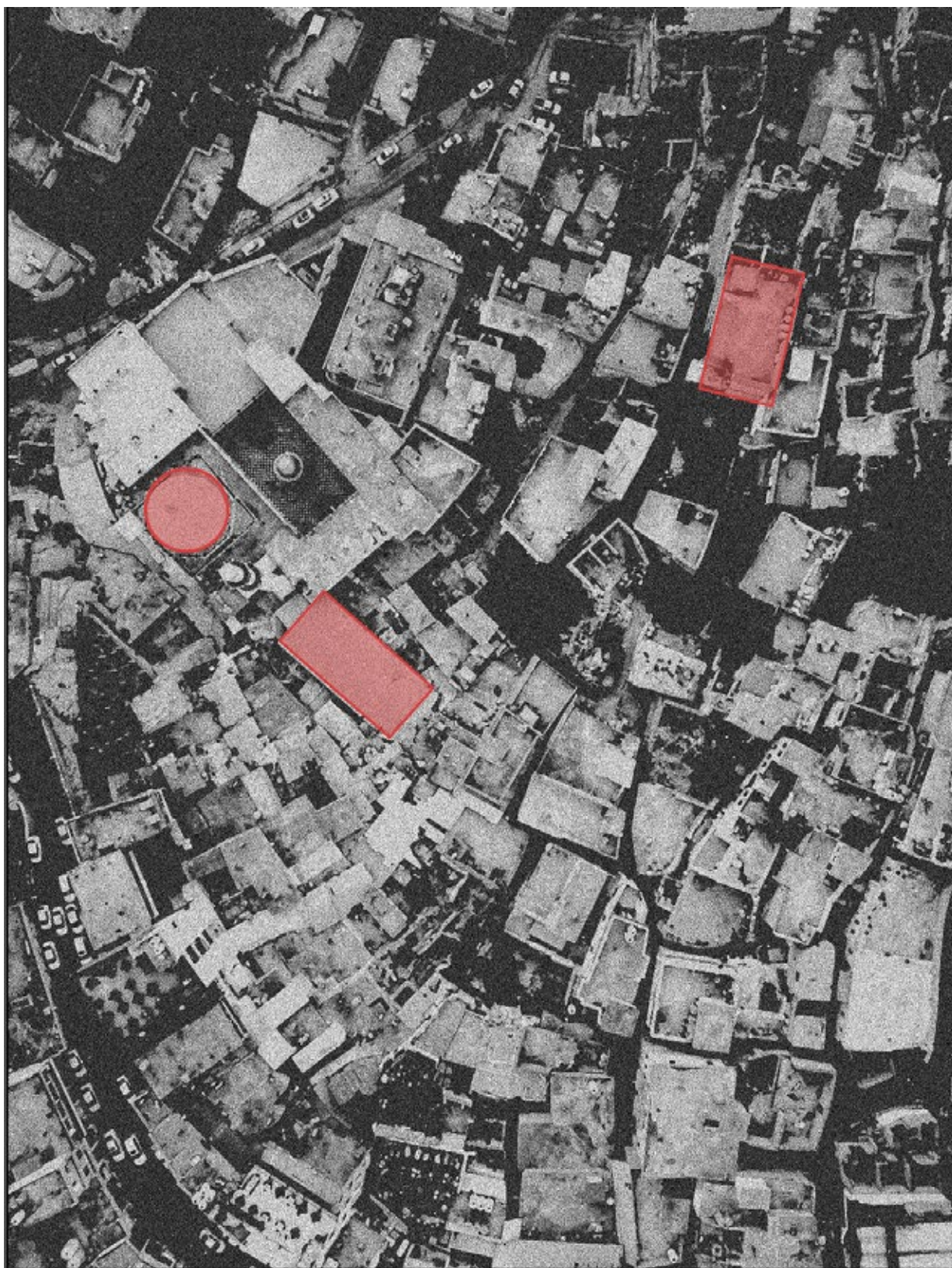
- **Baseline data:** building footprints used as a reference or data input can vary greatly. In May 2025, an updated OSM dataset contained 65% more buildings than the previously leading open Microsoft Building Footprints.
- **Methodology:** damage assessments can also differ in speed, accuracy, and imagery used.

Recommendations:

- **Damage data producers should share findings openly** on existing forums, stating intended use and limitations of the methodology and inputs.
- **No single damage assessment should be treated as fully accurate;** users should review limitations of damage assessments and consider the context of any damage findings.
- **Strengthen coordination and reduce duplication in damage mapping efforts** by dedicating resources to researching complementary strengths across different approaches.

**Bint Jbeil District, Southern Lebanon:
Total Damaged Buildings Identified from October - December 2024**





Aerial view of buildings in Beirut,
Lebanon. This image was captured
by iam hogir | [View on Pexels.](#)

BACKGROUND

Over the past five years, global conflict events have nearly doubled ([ACLED Conflict Index](#)), and understanding their impact on the ground is crucial for humanitarian planning, aid delivery, and early recovery. In this context, remote damage assessments rank among the top three satellite applications for humanitarians as they serve as an effective and safe approach for estimating infrastructure impacts ([Caribou: Beyond Borders](#)).

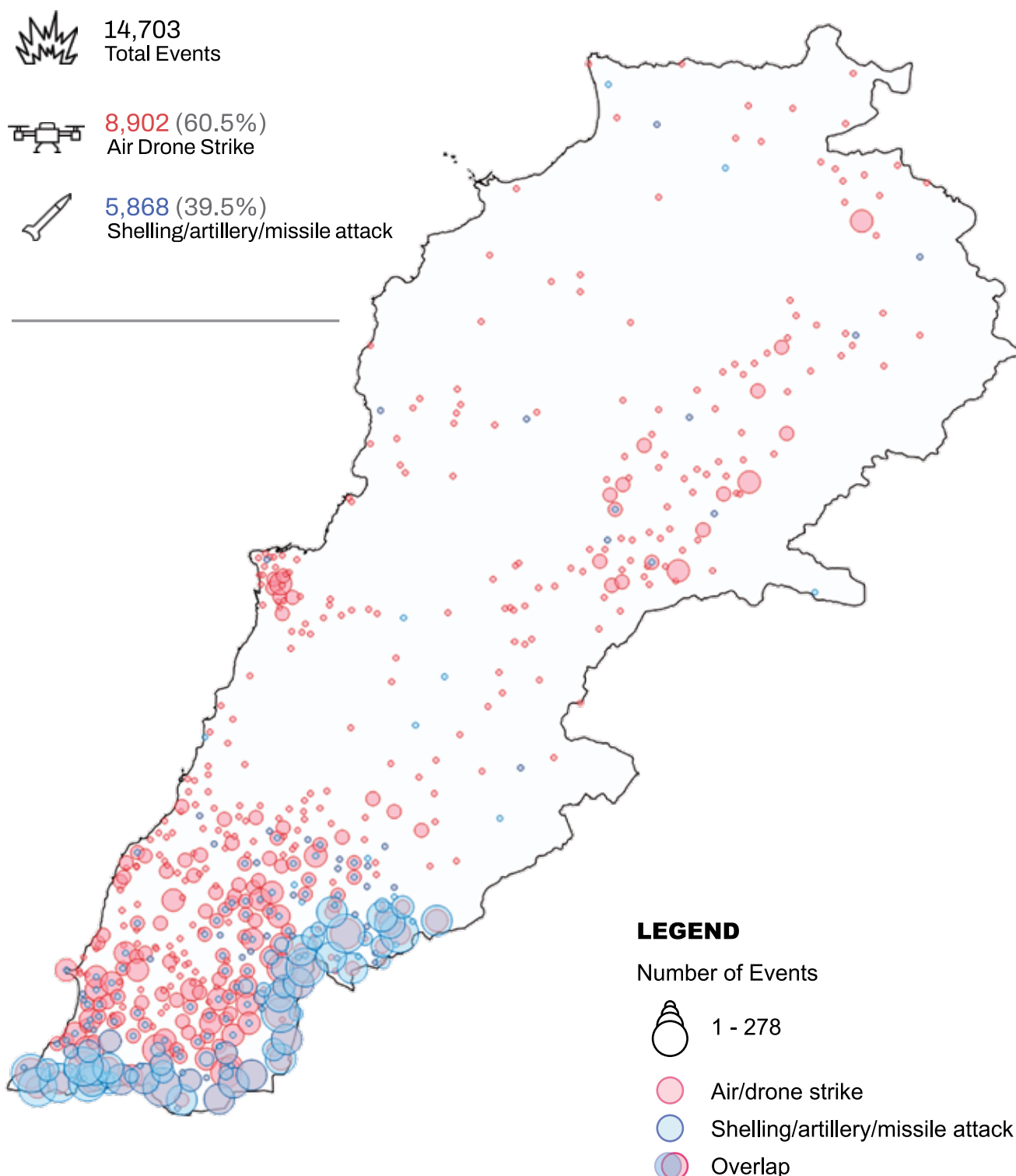
Using satellite imagery, damage can be tracked over time and across large areas, common aspects of conflict damage. In recent years, damage assessment methods have evolved, driven by advances in remote sensing technology and growing humanitarian needs in complex emergencies. For example, in recent conflicts like those in Ukraine and Gaza, limited access to ground information and widespread destruction led many who are researching conflict impacts to rely on remotely detected damage assessments ([Weekly building damage surveys in Gaza](#)). Open-access imagery, such as synthetic aperture radar (SAR) data from the [European Space Agency's Sentinel-1](#) constellation, also made damage analyses more accessible ([SAR in Support of the War in Ukraine](#)).

While having multiple lines of evidence of damage is important, large differences in findings can challenge confidence in remote findings.

New approaches to estimating building damages are continually being developed. With the prevalence of satellite-based damage assessments, it is important for end-users to understand the strengths and limitations of different approaches to mapping damage. While having multiple lines of evidence of damage is important, large differences in findings can challenge confidence in remote findings. Different results can cause misaligned programming across agencies. Also, for remote damage analyses to be operationally relevant, further collaboration and research are needed to develop a framework of recommended best practices for humanitarian use cases.

When conflict between Lebanon and Israel intensified throughout 2024 and sharply escalated in September 2024, multiple organizations began using remote damage detection methods. By the time a [ceasefire](#) was announced on November 27, 2024, at least one estimate predicted that up to 25% of the buildings along Lebanon's southern border had been damaged or destroyed ([Washington Post](#)). While the conflict in Lebanon looks to move towards early recovery, the impact of the conflict on buildings is of high interest, especially to those working on the ground in Lebanon ([Lebanon's Recovery: Mapping a Path from Rubble to Resilience](#)). However, damage assessments in Lebanon varied widely, prompting Humanitarian OpenStreetMap Team to investigate the approaches used by different organizations that are leading to such a wide spectrum of resulting damaged buildings estimates.

NUMBER OF **DRONE STRIKES** / **SHELLING** / **ARTILLERY** / **MISSILE ATTACKS** THAT OCCURRED IN LEBANON FROM OCTOBER 2023 TO DECEMBER 2024 AS COMPILED BY **ACLED** (ARMED CONFLICT LOCATION & EVENT DATA)



DIVERGENCE IN FINDINGS

A COMPARISON OF REMOTE DAMAGE ASSESSMENTS IN LEBANON 2024

Seven different analyses of damaged buildings in Lebanon were conducted as of October - December 2024 in response to the escalation in conflict, each with different findings, and summarized below.

Assessment	Org.	Imagery	Imagery Date	Publication Date	Remote Sensing Method	Damage	Building Footprints Source
Assessment of Damages in the South	MercyCorps	SAR	31 Oct 2024	2 Dec 2024	Backscatter differencing	"Damaged" "Destroyed"	Microsoft Buildings
The Extent of Building Damages in Lebanon	Miyamoto	SAR + optical	22 Nov 2024	16 Dec 2024	Mixed (Backscatter differencing + other)	"Moderate damage" "Severe damage"	Microsoft Buildings
Lebanon Damage Detection - Sentinel-2 L2A	ESRI Deep Learning	Optical	21 Oct 2024	28 Oct 2024	Automated with deep learning	"Damaged Structures"	Microsoft Buildings
Building destruction and debris quantities assessment	UN-Habitat, CREEMO	Optical	3 Dec 2024	22 Jan 2025	Expert visual interpretation	"totally and partially destroyed buildings"	OSM + Microsoft Buildings + custom data
Mapping building damage in Lebanon and northern Israel	Oregon State University	SAR	5 Dec 2024	8 Dec 2024	Coherent change detection	"Likely damaged or destroyed"	Microsoft Buildings
Lebanon Rapid Damage and Needs Assessment	WorldBank	SAR + optical	20 Dec 2024	March 2025	Mixed (Backscatter differencing + other)	NA - by sector analysis	OSM + Microsoft Buildings + custom data
From Crisis to Recovery: [...] Post-War Realities in Lebanon Rapid Impact Assessment	UNDP	NA - did not use earth observation/remote sensing	NA	22 Jan 2025	Primary data collection through direct communications.	NA - direct damage reports	NA

For the five assessments that released estimated building counts, the totals in Bint Jbeil District differed by a factor of three between two clusters:

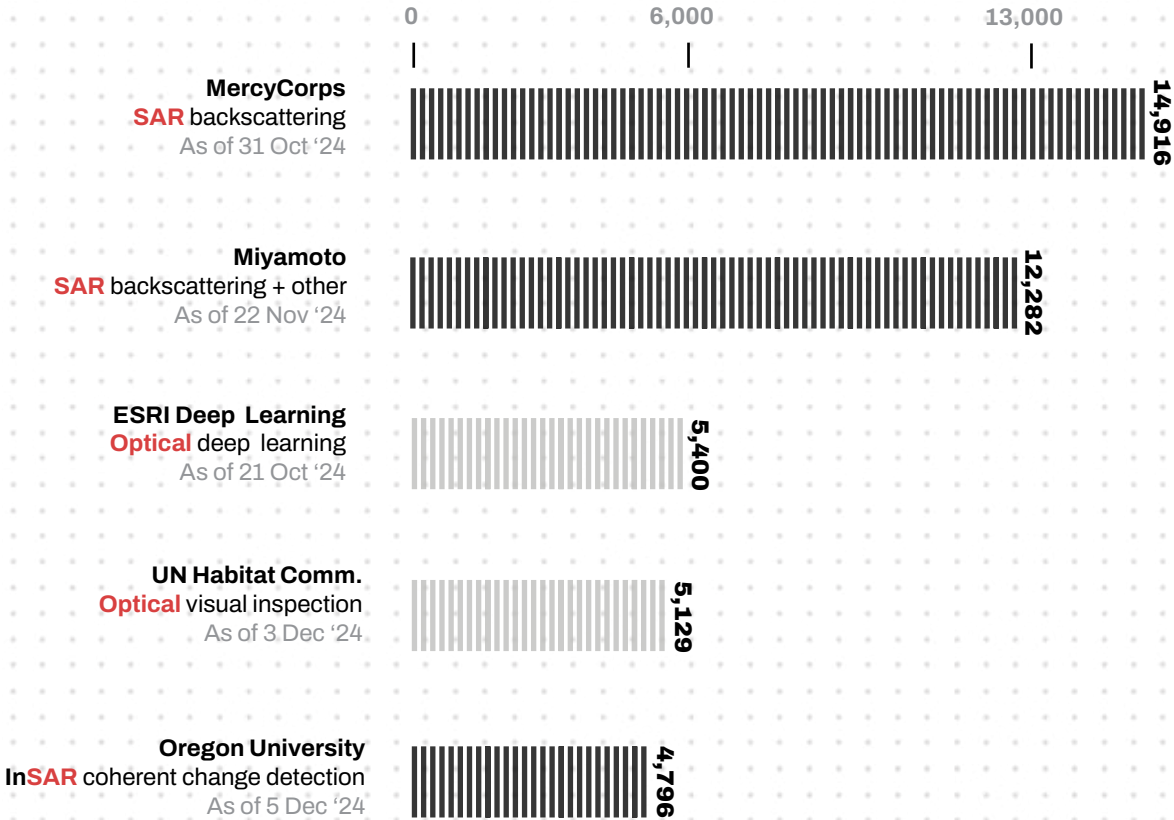
- Lower cluster: analyzed very high resolution optical imagery or InSAR coherent change detection.
- Higher cluster: analyzed backscatter intensity differencing with SAR.

Definitions of “damage” varied, but all the definitions indicated some level of damage to buildings during the conflict period. Time frames also differed slightly depending on the date of analysis and imagery.

However, differences in definitions and time frame do not appear to account for the massive difference shown. The lowest estimate (Oregon State University) covered one of the longest periods of time (as of 5 December 2024). Therefore, it can be assumed that the **discrepancies are due to differences in methodology and data inputs.**

The three lowest damage estimates used three different methodologies, indicating an emerging consensus around of 5,000 buildings being damaged in Bint Jbeil by October and November of 2024. Both of the higher estimates ($\approx 12\text{-}15,000$) were based on backscatter differencing from SAR imagery.

NUMBER OF DAMAGED STRUCTURES SAR AND OPTICAL IMAGERY



A spatial comparison of two SAR based methodologies in Bint Jbeil (see map 1) indicates a consistent pattern of higher concentration of damage along the southern border. However, the damage estimates varied significantly across the district, with some areas differing by 40%+. For example, there is high disagreement over the subdistrict of Rmeich, which was reportedly one of the only areas in Southern Lebanon not affected by Israeli military damage during this timeframe.

Such large differences in overall damage figures can undermine end-user confidence in remote sensing-based damage estimates, create confusion among decision makers, and potentially misdirect resources based on inaccurate damage

impact estimates. The number of different analyses in Lebanon makes it challenging to confidently cite the proportion of buildings damaged for advocacy, awareness, and resource allocation. On a programmatic level, knowing where to expect damage on the ground to plan for ground surveys and other response activities is challenging without a deeper understanding of the strengths and limitations of the type of remote damage methodology used in the analysis.

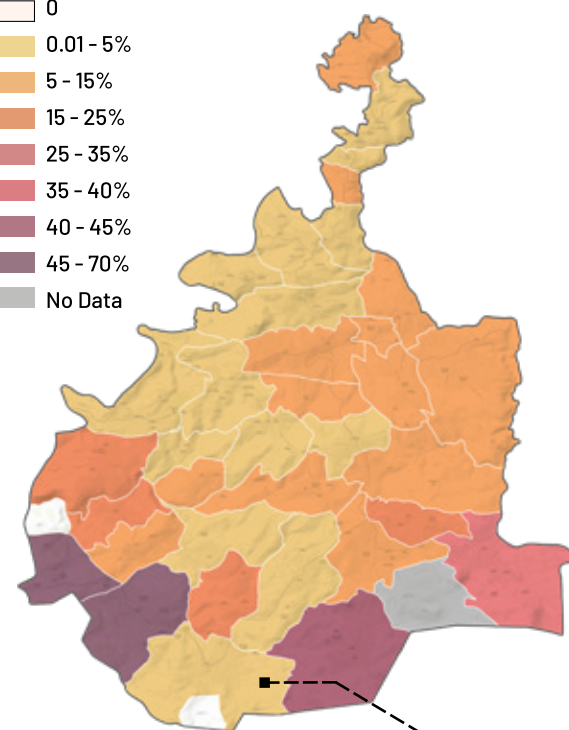
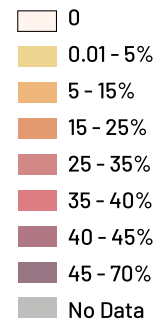
To understand what leads to large differences in damage assessments, the following sections are an overview of different remote damage analyses methods.

Map 1

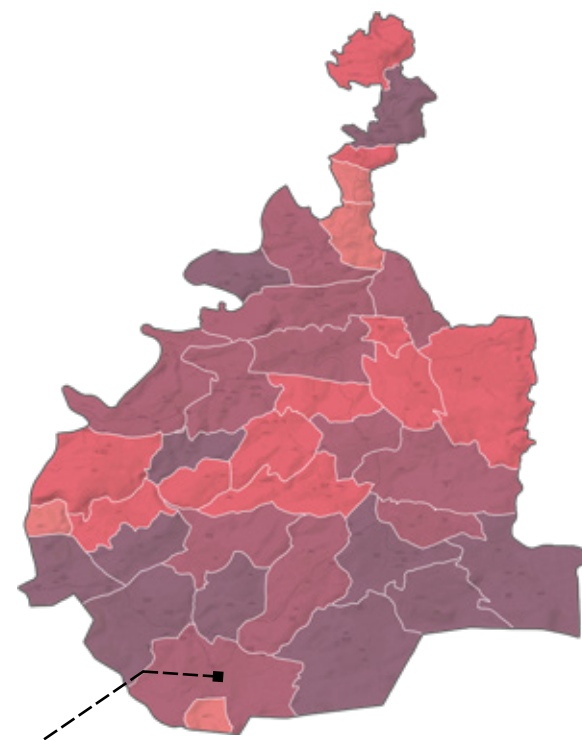
BINT JBEIL DISTRICT BUIDLINGS DAMAGED

LEGEND

Percent Estimate Damaged
Buildings



Oregon State University



MercyCorps

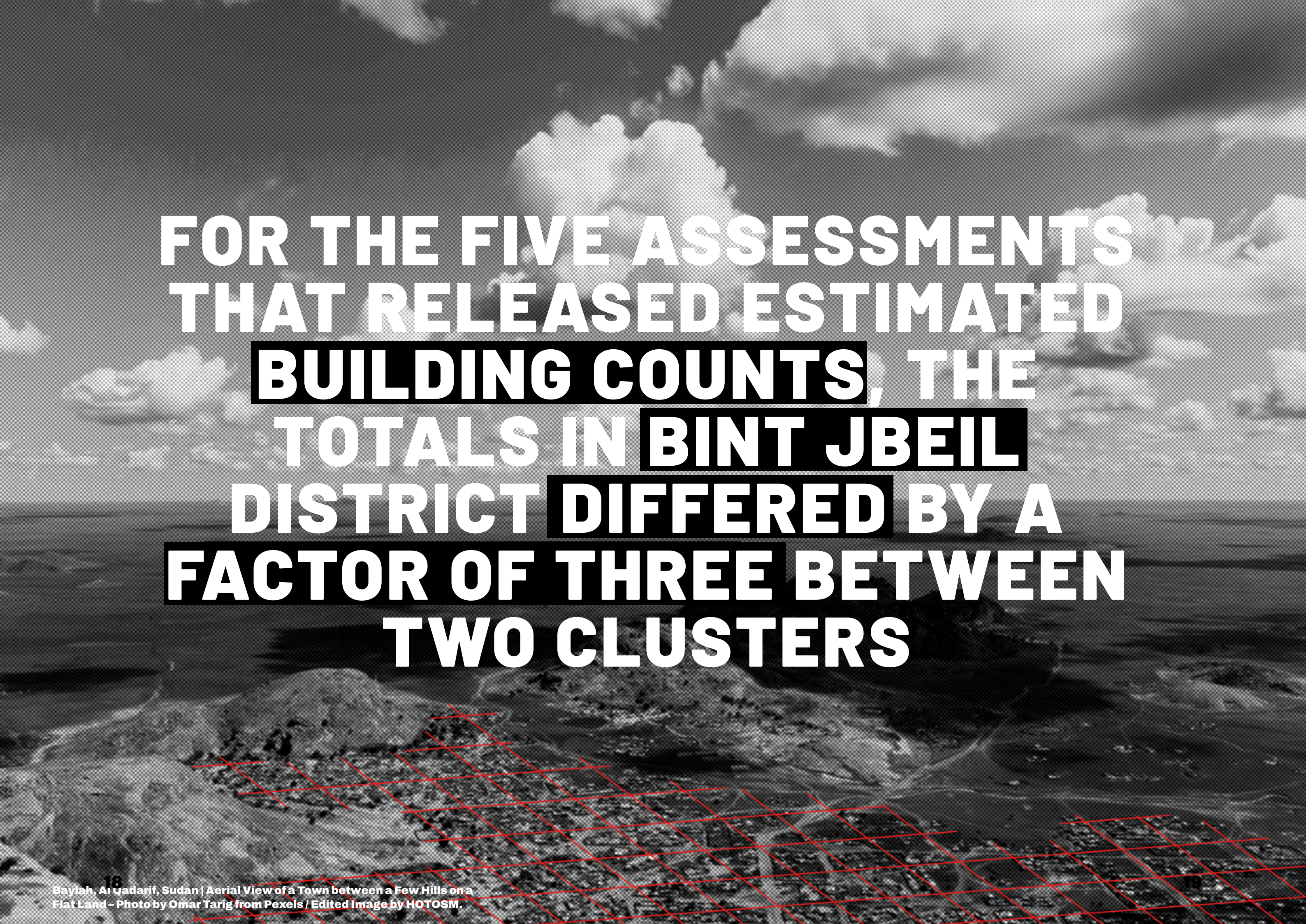
0.02% vs 40-45%

RMEICH

Imagery: ESRI Topographic basemap
Data: Assessment of Damages in the South (Mercy Corps), Mapping building damage in Lebanon and northern Israel (Oregon State University).

RMEICH , BINT JBEIL

[...] the damage estimates varied significantly across the district, with some areas differing by 40%+ damaged buildings between the two data sources. [...] there is high disagreement over the subdistrict of Rmeich [...] reportedly one of the only areas in Southern Lebanon not affected by Israeli military damage during this timeframe.

An aerial photograph of a town in Baylah, Al Qadarif, Sudan. The town is situated on a flat, arid landscape with some low hills. A red grid is overlaid on the ground, likely for mapping or assessment purposes. The sky is filled with large, white, billowing clouds.

**FOR THE FIVE ASSESSMENTS
THAT RELEASED ESTIMATED
BUILDING COUNTS, THE
TOTALS IN BINT JBEIL
DISTRICT DIFFERED BY A
FACTOR OF THREE BETWEEN
TWO CLUSTERS**

UNDERSTANDING DIFFERENT APPROACHES FOR CONFLICT DAMAGE MAPPING

Regardless of the different approaches to remote sensing analysis of conflict damage and their varying results, all share common workflows that rely on key inputs. In this section, specific steps in remote sensing damage workflows are explored to inform end-users of the mechanisms behind certain results. The main components of remote damage assessment workflows are the following (not necessarily done in this order): 1) a pre-conflict baseline building footprints dataset, 2) defining of parameters for damage, and 3) remote sensing methods.

1. Building Footprints As Pre-Conflict Baseline

Building footprint data can be used as input data for remote damage analyses or provide spatial context for an assessment. In the first case, image values within building footprints may be directly used to assess damage to a location; in the second case, footprints may not be considered in the damage detection itself but may be used to aggregate ‘damaged’ locations or regions to estimate the number of likely damaged buildings. Complete and precise building footprints

are critical to ensure confidence in identified building damage, as well as **estimates of the proportion of buildings that are likely damaged**. One of the most important factors for aligning different damage assessments is the consistent use of the same high-quality baseline building data. Most of the damage assessments reviewed in this report used Microsoft Building Footprints, which at end of 2024 had the best buildings coverage available when most of the assessments were conducted.

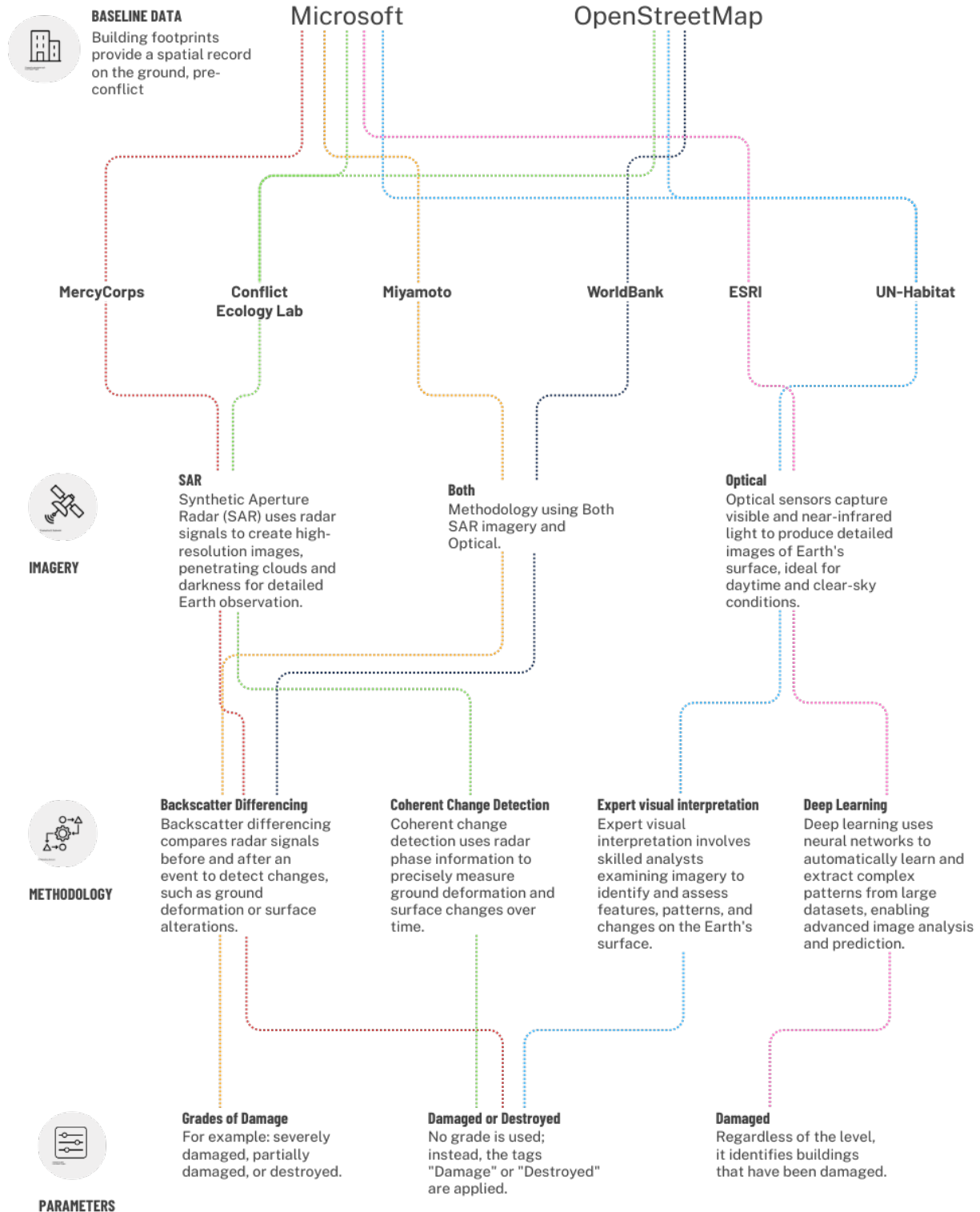
In March 2025, a [crowdsourced volunteer effort](#) coordinated by Humanitarian OpenStreetMap Team to update building footprints in conflict affected areas of Lebanon was completed. As of May 2025, a comparison between OSM and available building footprints in Lebanon’s southernmost conflict affected districts found OpenStreetMap (OSM) buildings had 65% more buildings than Microsoft, and more buildings than Overture, which is an aggregate of both OSM and Microsoft (see [table 1](#)).

District	Overture	Microsoft Buildings	OSM Buildings	% OSM > MS
Bint Jbeil	36,730	31,176	48,470	+55.5%
Maarjyoun	30,889	28,647	39,462	+37.8%
Nabatiyeh	53,739	45,365	78,388	+72.8%
Sour	70,393	60,515	87,344	+44.3%

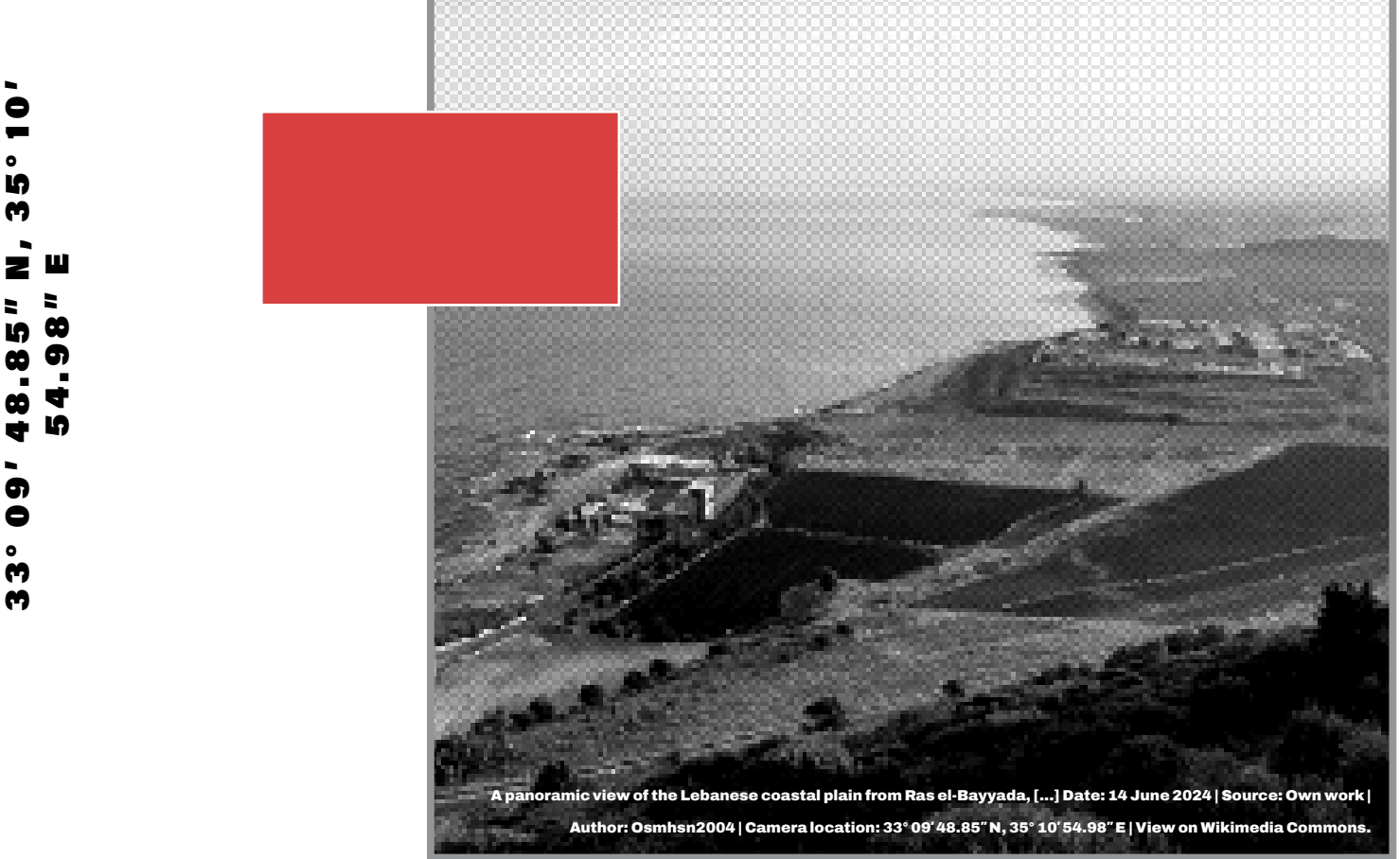
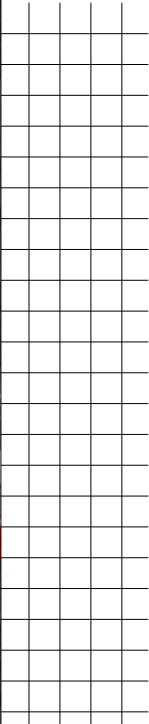
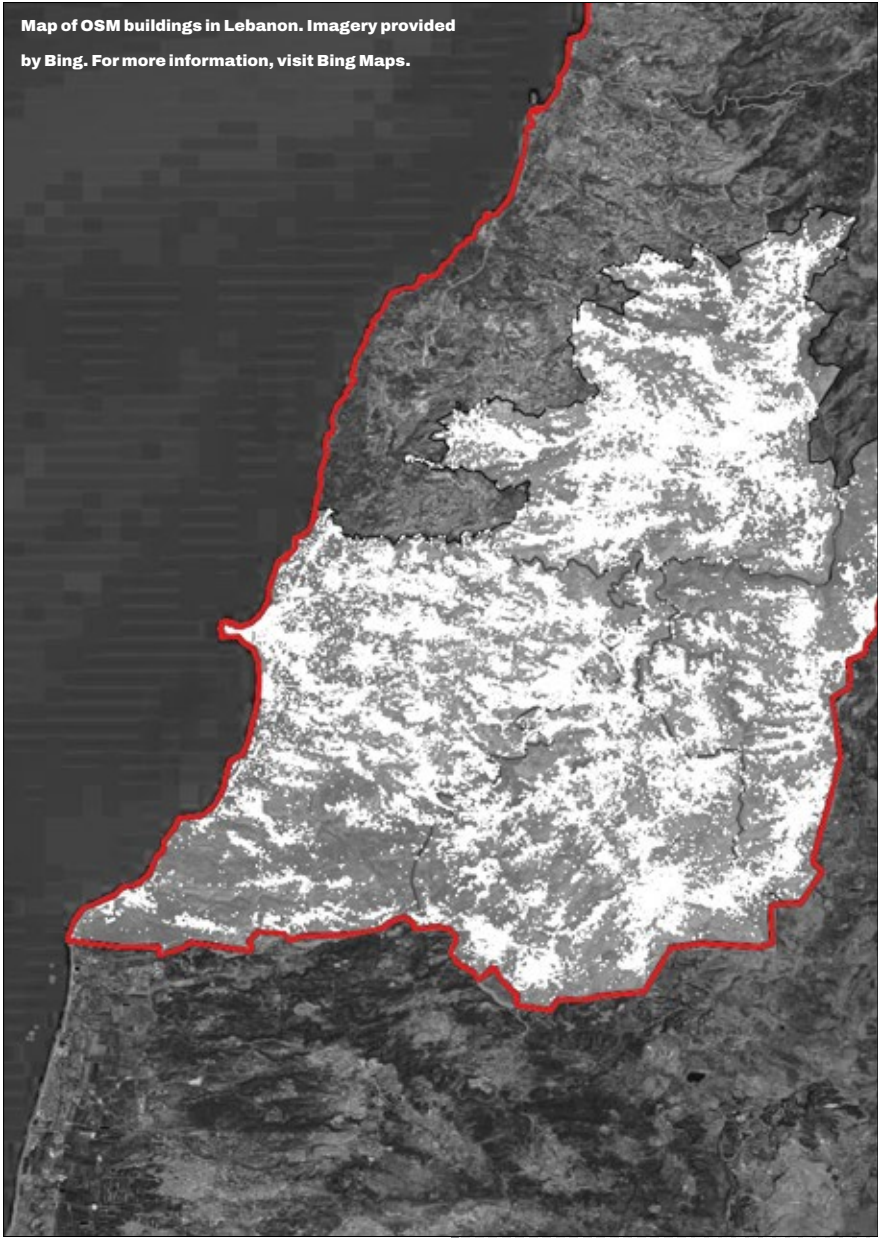
Table 1. Comparison of building datasets from OpenStreetMap, Microsoft, and Overture in Southern Lebanon. The “% OSM > MS” column shows how much larger the OSM dataset is compared to Microsoft’s dataset.

WORKFLOWS

+ PARAMETERS



Map of OSM buildings in Lebanon. Imagery provided by Bing. For more information, visit Bing Maps.



33° 09' 48.85" N, 35° 10' 54.98" E



A panoramic view of the Lebanese coastal plain from Ras el-Bayyada, [...] Date: 14 June 2024 | Source: Own work | Author: Osmhsn2004 | Camera location: 33° 09' 48.85" N, 35° 10' 54.98" E | View on Wikimedia Commons.

As the need for timely damage estimates outweighed concerns of missing buildings in available data sources, the damage assessments conducted at the end of 2024 possibly overlooked damage in up to 65% of buildings that were absent from the existing datasets, buildings that have since been added by crowdsourcing to OSM.

The change in the best source for building footprints from Microsoft to OSM indicates that those using building datasets for conflict damage assessments should monitor coverage carefully before beginning their analysis, as the best source can change over time.

2. Establishing Parameters for Damage

There are two main parameters to define for any approach to mapping/digitizing conflict damage: the time period of the analysis, and the categorization(s) of damage.

Establishing the period of analysis will depend on the availability of the imagery used. However, as mentioned in the background section, the time frame did not seem to greatly influence results during this period in Lebanon, given that the assessment covering the longest period of time had the lowest estimate for damage.

In addition to defining time period, there are two main approaches to classifying damage in structures:

- **Ranked Classification:** Assigns a specific damage level to each structure, such as “moderate,” “severe,” or “destroyed.” In Southern Lebanon, two assessments used this approach: Miyamoto and MercyCorps, which had the highest counts for damage.
- **Binary Classification:** Simply indicates whether a structure is damaged, regardless of severity. In this case, assessments by Oregon State University, ESRI, and UN-Habitat used this style of classification.

Further studies are needed to explore the relationship between the use of these classification systems and the outcomes of the assessment.

3. Remote Sensing Methods

Remote sensing methods used to detect damage depend on the type of aerial imagery used: optical (including true color images) or Synthetic Aperture Radar (SAR). Each type of imagery have two common remote damage detection methods with limitations and benefits. All the common methods are described in the infographic on pages 14-15 and **table 2**.

Methodology	Quick Can be completed in < 2 days with preparation	Can Use Available Open-Access Imagery	Open Process Can run using open source applications
Optical - deep learning	✓		
Optical - expert visual interpretation			✓
SAR* - coherent change detection	✓	✓	
SAR* - backscatter differencing	✓	✓	✓

*Sentinel-1 open imagery

Table 2. Accessibility comparison of different remote sensing analysis approaches

SATELLITE IMAGERY

OPTICAL IMAGERY

Optical imagery employs **visible light**, to generate images, relying on reflected sunlight for this process. Because of this it is suitable for **humans to use for visual inspection**. Optical imagery can be hindered by factors such as viewing angle, weather conditions, or low resolution. Viewing angle tends to be from top.

Optical and SAR imagery differ in their acquisition methods. **Optical imagery relies on visible light**, making it susceptible to weather and lighting conditions, while **SAR uses microwave signals**, enabling all-weather and nighttime imaging.

OPTICAL TECHNIQUES

Manual Detection

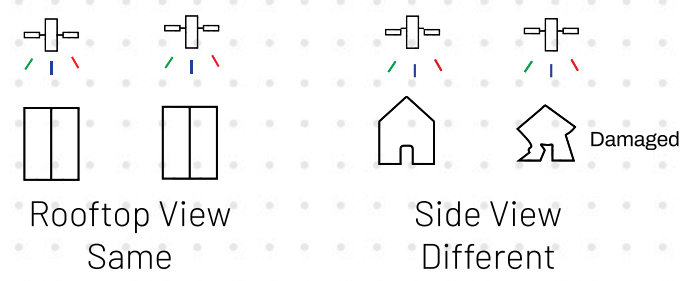
Even at high resolution, manual damage detection may struggle with depth and detail, limiting effectiveness in identifying subtle structural changes and might require a lot of trained people.

Automated Detection

Techniques such as machine learning and image processing enhance the system's ability to identify damage. Although this eases limitations related to resolution and angle, they still apply.

LIMITATIONS

Details on less extensive damage can be concealed from optical imagery due to angle or resolution.



SAR IMAGERY

Synthetic Aperture Radar (SAR) uses **radio and micro waves** signals of varying wavelengths to create images of the Earth's surface. SAR can penetrate clouds, canopies, objects and darkness – the longer the wave, the more penetrating. The slanted angle of the satellites improves its detection capabilities.

SAR TECHNIQUES

Backscatter

- Uses Amplitude only

Amplitude measures the **intensity of returned signal** echoes. Variations in intensity can be influenced by surface roughness and material properties, and be used to detect full collapse of structures

Coherence Change

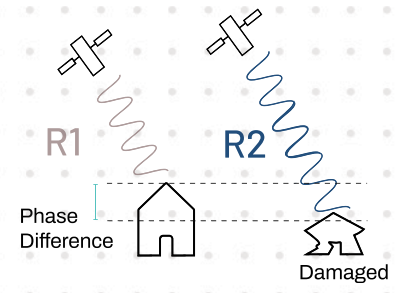
- Uses Amplitude and Phase

This technique measures **changes in both amplitude (backscatter) and phase**, allowing for the calculation of coherence, which is a measure of the correlation between the phase of two or more SAR images (InSAR).

Interferometric SAR (InSAR)

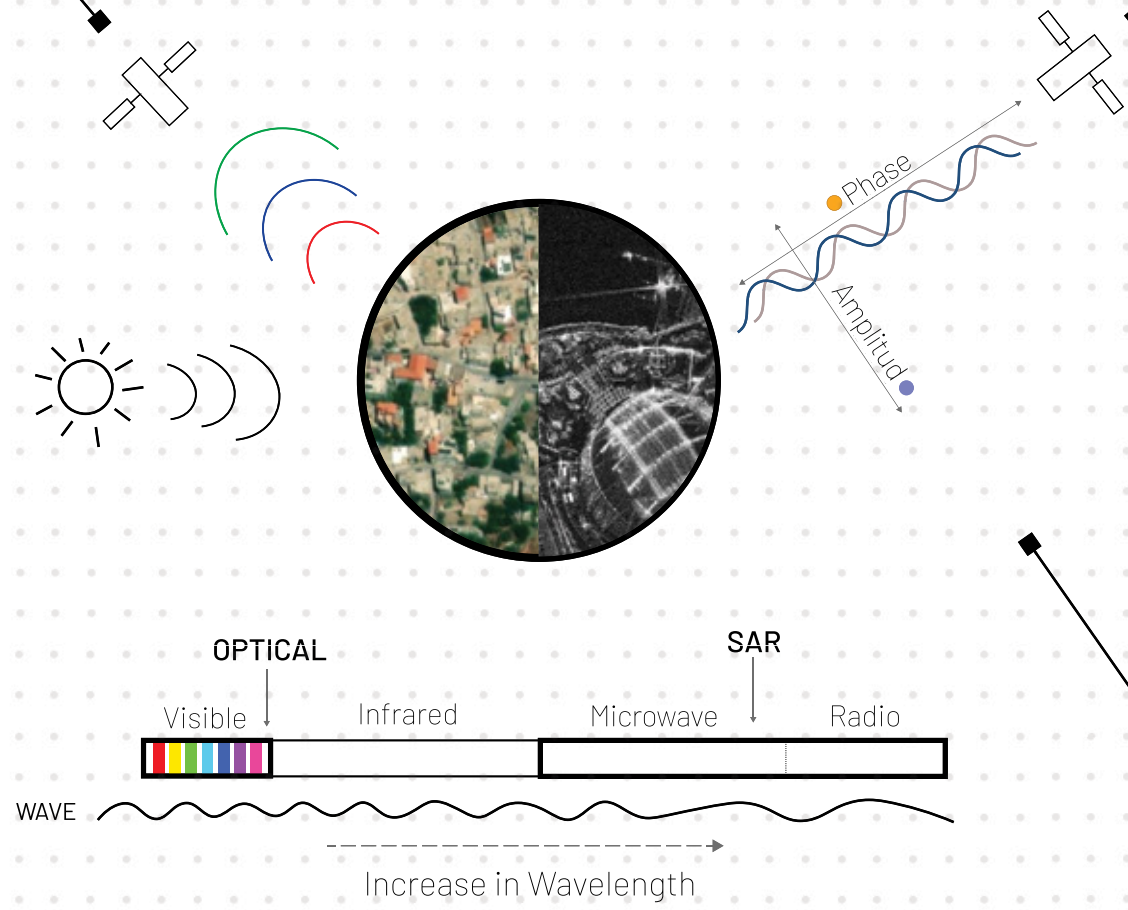
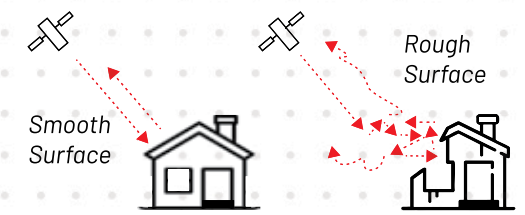
- PHASE DIFFERENCE

InSAR is the measurement of the phase difference between two radar images (R1 and R2) taken at two distinct times over the same area. This increases the likelihood of detecting changes even if subtle.



- AMPLITUDE

Schematic sketch with example of scattering SAR types¹



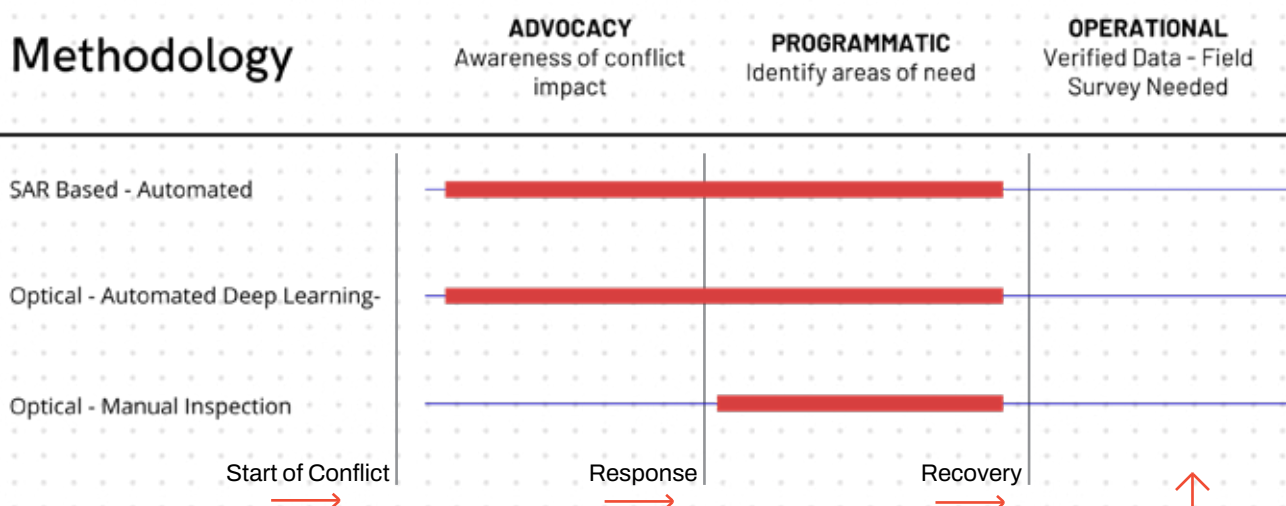
DISCUSSION

A thorough understanding of conflict damage across affected areas requires considering multiple approaches ([Remote Sensing Comparisons Case Study](#)). For this reason, establishing multiple lines of evidence for damage to infrastructure is useful to build a confident narrative of conflict impact on communities and understand the full consequences of the damage.

In practice, different damage analysis methods vary in the time they take and the satellite imagery they require. **Understanding the reasons behind these differences is important for determining when it is appropriate to use a particular assessment for responsible data use.** Rapid assessments, while potentially incomplete (e.g. due to missing building footprints) are intended to provide general estimates and trends. Rapid assessments can still support early advocacy-oriented messaging, provided the expertise behind the estimate is confirmed and updated as more information becomes available.

With more time, damage assessments become more refined with better data and can be used to inform more programmatic and operational planning. It is also important to consider how different types of damage analysis can inform one another. More timely remote damage analyses can be used to advocate for aid response and recovery resources, as well as inform more detailed damage surveys or ground operations of aid delivery.

Humanitarian Applications Of Remote Damage Conflict Damage



Damage assessments suitable for operational use are recommended to only come from ground-truthed and verified surveys. This requires a significant increase in resource investment, but it is the minimum necessary to avoid potential backlash when presenting such sensitive data and to maximize the effective allocation of resources for post-conflict and disaster efforts.

An aerial photograph of a densely populated urban area in Lebanon. Overlaid on the grayscale satellite imagery are numerous black-outlined polygons representing building footprints from the OpenStreetMap (OSM) database. The buildings vary in size and shape, mostly rectangular, and are packed closely together. A road network is visible, with some wider thoroughfares and narrower alleys. The text 'RAPID ASSESSMENTS, EVEN IF INCOMPLETE, CAN SUPPORT ADVOCACY IF EXPERTISE IS CONFIRMED.' is superimposed on the left side of the image, enclosed in a white box with a black border.

**RAPID ASSESSMENTS,
EVEN IF INCOMPLETE,
CAN SUPPORT
ADVOCACY IF
EXPERTISE IS
CONFIRMED.**

Close up of OSM buildings in Lebanon. Imagery provided by Bing. For more information, visit [Bing Maps](#).

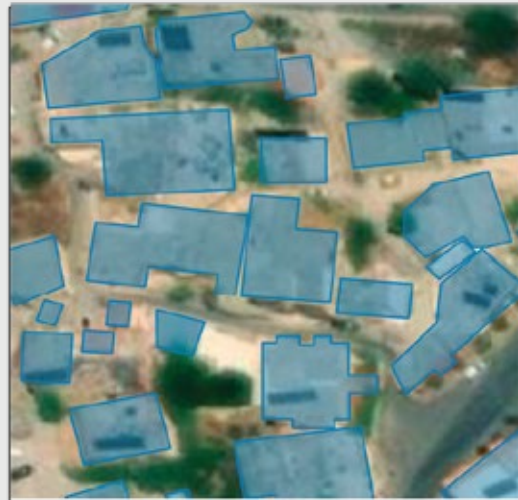
BUILDING FOOTPRINT DATA COMPARISON IN SOUTHERN LEBANON

BUILDING GEOMETRY

Microsoft ML Oct. 2024



OSM March 2025



COMPLETNESS



POSITIONAL ACCURACY



Imagery sourced from © Google and © Esri and its licensors.

A side by side comparison of building footprints from two sources, Microsoft and OpenStreetMap, comparing for position, geometry and completeness.

CONCLUSION

In the current times of reduced resources for humanitarian organizations, complementary and coordinated damage analyses are needed. While multiple lines of evidence for damage from different approaches can create a better informed response, misinterpretation and widely different findings can confuse and undermine the use of remote damage assessment data.

Better research and coordination between damage data producers and intended audiences are needed for:

- Full transparency from damage data producers on the extent of analysis, the methodology on how the data was produced, what the data should be used for, and the methodological limitations.
- Open access to damage data (potentially aggregated) for responsible application to multiple use cases, to avoid unnecessary duplication due to a lack of access.

- Coordination of damage assessments to ensure complementary and responsible use of damage data across organizations to:

- **Leverage the entire damage assessment ecosystem.**
- **Establish best practices for when to use specific damage assessment methodologies.**
- **Increase the use of accepted vetted damage analyses and reduce duplication.**

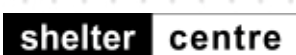
- Quality open building footprints: Invest in OpenStreetMap communities to ensure timely, comprehensive, human-in-the-loop building footprints are in place for use by damage assessments.

More research is also needed to make damage analyses more operationally relevant. Multiple methods of damage analyses, such as InSAR coherent change detection and visual interpretation, can inform each other, and dedicated resources are needed to harmonize analyses and inform humanitarian decision-making.

ACKNOWLEDGMENTS



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